International School

Capstone Project 2

CMU-IS 451

Project Proposal

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MentorAI Grad – Intelligent Thesis & Capstone Project Management System

Submitted by C2SE.71

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PROJECT INFORMATION

Project acronym	MAG		
Project Title	MentorAI Grad – Intelligent Thesis & Capstone Project Management System.		
Start Date	1 Feb 2026	**End Date**	11 May 2026
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TABLE OF CONTENTS

1. Project Title 2
2. Team Members 2
3. Supervisor(s) 2
4. Problem Statement 2
5. Survey / Existing Solutions 4
6. Objectives and Scope 5
    6.1. Objectives 5
    6.2. Scope 5
7. Key Feature & Requirement 7
    7.1. Key Features 7
    7.2. Functional Requirement 8
    7.3. Non-Function Requirement 9
8. Constraints and Assumptions 9
    8.1. Constraints 9
    8.2. Assumptions 10
9. Target User / Stakeholder 10
    9.1. Target User 10
    9.2. Stakeholder 11
10. Technology Stack 11
11. Methodology & Development Plan 12
    11.1. Methodology 12
    11.2. Development Plan 13
12. System Architecture Overview 14
    12.1. System Context Diagram 14
    12.2. System Context Description 14
13. Potential Risks and Mitigation Strategies 15
14. Expected Outcomes / Deliverables 17
15. Project Constraint 18
16. References 19

LIST OF TABLES

Table 01. *Team Members* 2
Table 02. *Supervisor* 2
Table 03. *Survey / Existing Solutions* 4
Table 04. *Key Features* 7
Table 05. *Functional Requirements (FR)* 8
Table 06. *Non-Functional Requirements (NFR)* 9
Table 07. *Target User* 10
Table 08. *Stakeholder* 11
Table 09. *Timeline* 13
Table 10. *Project team* 13
Table 11. *Potential Risks and Mitigation Strategies* 15
Table 12. *Project Constraint* 18
Table 13. *References* 19

LIST OF FIGURES

Figure 01. *Scrum Process* 12

Figure 02. *System Context Diagram* 14

1. Project Title

MentorAI Grad – Intelligent Thesis & Capstone Project Management System.

2. Team Members

Table 01. *Team Members*

Name	Student ID	Email	Role
Tu Pham Thi Suong Tuyet	28201152213	tuphamsuongtuyet@gmail.com	Scrum master , Frontend Dev, Backend Dev.
Tran Dai Nghia	27211243595	Trandainghia3703@gmail.com	Frontend Dev, Backend Dev.
Tran Van Luc	28211105758	vanluccbr@gmail.com	Frontend Dev, Backend Dev.
Nguyen Thi Nga	28207144919	nguyenthinga46204@gmail.com	Frontend Dev, Backend Dev.
Tran Anh Thai	28211152669	henny4gaming@gmail.com	Frontend Dev, Backend Dev.

3. Supervisor(s)

Table 02. *Supervisor*

Name	Department	Email
Nguyen Trung Thuan, MSc	E-Learning	nguyentrungthuan@gmail.com

4. Problem Statement

4.1. What difficulties are users/customers facing?

For Students:

Students face many difficulties during the thesis/project implementation process.

- Project requirements and progress tracking are often unclear, as communication mainly relies on emails, chat messages, or verbal instructions.
- Lecturer feedback is usually long and detailed, making it difficult for students to:
  + Identify key issues that need to be fixed.
  + Prioritize tasks for the next milestone.
- Students lack a centralized system to:
  + View project information, milestones, and deadlines in one place.
  + Track submission history and lecturer feedback.
- There is no early warning mechanism to notify students of potential delays based on:
  + Missed deadlines.
  + Late submissions in previous milestones.

- Students do not receive structured guidance on how to plan and complete the next milestone efficiently.

For Lecturers

Lecturers also encounter significant challenges in managing student projects:

- Project management is often manual and fragmented, using spreadsheets, emails, or messaging apps.
- Tracking the progress of multiple students or groups is time-consuming and inefficient.
- Reviewing submissions and writing feedback for each milestone requires substantial effort.
- There is no system support to:
  + Quickly summarize student progress.
  + Identify students or groups at risk of falling behind schedule.
- Lack of centralized data makes it difficult to evaluate overall project status and ensure consistent supervision.

4.2. What is the impact of this problem?

- Students may misunderstand lecturer expectations, leading to repeated revisions and lower project quality.
- Missed or delayed milestones increase stress and reduce students' chances of completing their projects on time.
- Lecturers spend excessive time on administrative and tracking tasks instead of focusing on academic guidance.
- Delayed feedback cycles negatively affect learning outcomes and project improvement.
- The overall project supervision process becomes inefficient due to:
  + Lack of centralized project data.
  + No automated progress monitoring.
  + Poor visibility of risks and delays.
- Both students and lecturers waste time and effort because there is no intelligent system to support structured project management.

4.3. Why did the team choose to solve this problem?

The team chose to solve this problem because the current thesis and project supervision process is fragmented, unclear, and inefficient for both students and lecturers. Students struggle with unclear requirements, scattered feedback, and the lack of a centralized system to track progress, milestones, and deadlines, which increases the risk of delays and repeated revisions. Lecturers, on the other hand, face difficulties managing multiple projects due to manual tracking methods and the absence of tools to quickly monitor progress or identify at-risk students. These issues lead to inefficient feedback cycles, increased workload, and reduced learning outcomes. Moreover, the lack of automated progress monitoring and early warning mechanisms negatively affects project quality and timely completion. Therefore, the team aims to develop a centralized and intelligent system to improve project supervision efficiency, transparency, and overall academic outcomes.

5. Survey / Existing Solutions

Table 03. *Survey / Existing Solutions*

Characteristics	MentorAI Grad	Trello	Notion	Jira	GitHub Projects
Primary Purpose	Capstone Project Management	Task Management	Multipurpose Workspace	Issue & Sprint Management	Code-Linked Task Management
Target Users	Students, Instructors	General teams	Individuals & teams	Software teams	IT students & developers
Research Topic Approval	✓ There is a process for reviewing and assigning instructors.	x	x	x	x
Academic Workflow Support	✓ Predefined phases (Proposal → Midterm → Final)	x	x	x	x
Instructor Feedback	✓ Structured feedback per milestone	Comments only	Comments only	Comments only	Comments only

Summarizing feedback using AI.	✓	x	Content writing	Task generation	x
Warning of potential delays	✓	x	x	Deadline alerts	x

6. Objectives and Scope

6.1. Objectives

The overall objective of this project is to design and develop a web-based system that supports the management and monitoring of student capstone projects, integrating AI-based features to assist students and supervisors in tracking progress, improving communication, and reducing project delays.

Specific Objectives (SMART)

SMART: Specific, Measurable, Attainable, Relevant, and Time-bound

- Specific: To develop a web-based capstone project management system that supports project topic management, milestone tracking, submissions, and supervisor feedback within a single platform.
- Measurable: To implement milestone-based progress tracking that allows supervisors to monitor student progress using clear status indicators (on-time, submitted, late).
- Attainable: To provide a structured submission and feedback mechanism that can be realistically implemented within the project timeline and available technical resources.
- Relevant: To address real-world challenges in supervising and managing capstone projects for both students and supervisors in an academic environment.
- Time-bound: To complete the design, development, and deployment of the system within the duration of the Capstone Project course.

6.2. Scope

6.2.1 In-Scope (Within the Scope of the Project)

The scope of this project focuses on designing and developing a web-based capstone project management system as a pilot Minimum Viable Product (MVP) for a limited group of Capstone Project students and supervisors within the academic environment of Duy Tan University, specifically targeting student groups participating in the Capstone Project 2 course.

The system will include the following core functionalities:

For Students

- User registration, login, and profile management
- Project topic registration and management
- Creation and update of project execution plans
- Submission of reports and documents based on milestones
- Viewing supervisor feedback and comments
- Receiving notifications and deadline reminders
- Viewing and analyzing personal project progress
- Performing self-assessment of project progress

For Supervisors

- User authentication and profile management
- Management of supervised students and project topics
- Approval and assignment of project topics
- Monitoring progress of multiple students or groups
- Reviewing submissions and providing feedback
- Viewing dashboard for overall supervision status
- Receiving alerts for students at risk of missing deadlines
- Managing meeting schedules with students

AI-Supported Features

- AI-assisted summarization of lecturer feedback and student reports
- (Optional) AI chatbot for FAQ using predefined prompts

User Scope

- Undergraduate students working on capstone or final-year projects
- Supervisors (lecturers) responsible for guiding and evaluating students

Technical Scope

- Web-based application accessible via modern browsers
- Centralized database for managing projects, milestones, and submissions
- Integration of AI services for analysis and assistance
- Role-based access control for different user types

6.2.2 Out-of-Scope (Excluded from the Project)

The following items are explicitly excluded from the scope of this project:

- Automatic grading or final evaluation of student projects by AI
- Replacement of supervisors' decisions or academic judgment
- Integration with external university systems (e.g., LMS, student information systems)
- Real-time plagiarism detection or similarity checking
- Mobile native application development (Android/iOS)
- Financial, payment, or administrative management features
- Large-scale deployment for multiple universities

6.2.3 Scope Justification

The scope is intentionally limited to ensure:

- Feasibility within the project timeline
- Alignment with academic requirements of the course
- Clear focus on core project management and supervision challenges

Effective demonstration of AI as a supporting tool, not a replacement for human decision-making

7. Key Feature & Requirement

7.1. Key Features

Table 04. *Key Features*

No.	Feature	Description
1	User & Role Management	Allows users to register, log in, and manage profiles with role-based access (Student, Supervisor, Admin).
2	Project Registration & Approval	Enables students to register project topics and supervisors to review, approve, or assign them.
3	Progress Planning & Tracking	Allows students to create and update project plans, upload progress reports, and track milestones.
4	Communication & Collaboration	Facilitates messaging, feedback, and scheduling meetings between students and supervisors.
5	Notification & Alert System	Sends automated reminders for deadlines and alerts for at-risk or delayed progress.
6	AI-Assisted Feedback and Report Summarization	The system integrates an AI service to generate concise summaries from long lecturer feedback and student reports. This feature helps students quickly identify key issues and required improvements. The AI functionality is implemented using prompt engineering with the Gemini API and does not involve custom model training

		or predictive analysis.
7	Dashboard & Reporting	Displays personalized dashboards for students and supervisors, and provides system-wide reports for administrators.
8	Administration & System Control	Allows administrators to manage users, configure system settings, monitor activities, and export system-wide reports.

7.2. Functional Requirement

Table 05. *Functional Requirements (FR)*

ID	Requirement	Description
FR01	Manage Users & Roles	The system must allow users to register, log in, log out and manage personal information with access by role (Student, Instructor, Admin).
FR02	Registration & Project Approval	The system must allow students to submit research topics and for faculty members to review, approve, or assign them.
FR03	Planning and Tracking Progress	The system must allow students to create, update, and track project plans, and upload progress reports with version control.
FR04	Communication & Collaboration	The system must support real-time messaging, feedback exchange, and meeting scheduling between students and faculty.
FR05	Notification & Alert System	The system must automatically send deadline reminders and generate alerts for progress that is at risk or behind schedule.
FR06	Dashboard & Reports	The system must provide a personalized dashboard for students and faculty, displaying progress, scores, and outstanding tasks.
FR07	AI-Assisted Summarization	The system shall provide an AI-assisted feature to summarize textual feedback and reports to support students' understanding of lecturer comments. The AI service operates based on predefined prompts using the Gemini API (free tier) and is limited to a fixed number of requests per minute.
FR08	System Administration	The system must allow administrators to manage user accounts, configure the system, monitor activity logs, and generate system-wide reports.
FR09	File & Document Management	The system must support uploading, storing, and organizing project-related documents (reports, plans, feedback) with access control.
FR10	Assessment & Feedback	The system must allow instructors to assess student progress, provide graded feedback, and track performance history.

7.3. Non-Function Requirement

Table 06. *Non-Functional Requirements (NFR)*

ID	Requirement	Description
NFR01	Performance	The system shall respond to user requests within 3 seconds under normal operating conditions.
NFR02	Security	User passwords are securely stored using bcrypt hashing. Role-based access control ensures that students can only access their own projects and documents. File uploads are restricted to PDF and DOCX formats with a defined size limit (e.g., 10MB), and access controls are applied to prevent unauthorized file downloads.
NFR03	Usability	The system interface shall be intuitive and easy to use for users with basic technical skills.
NFR04	Availability	The system shall maintain an availability level of at least 95% uptime.
NFR05	Compatibility	The web application shall be compatible with modern desktop and mobile web browsers.
NFR06	Scalability	The system shall be designed to support an increasing number of users without major architectural changes.

8. Constraints and Assumptions

8.1. Constraints

Time Constraints:

- The system is developed within a limited timeframe from February 1st to May 11th.
- Due to time limitations, the project focuses on core capstone project management features and basic AI-assisted support.
- Advanced features such as mobile applications or complex AI models are excluded from the project scope.

Technology Constraints:

- AI-assisted features rely on existing APIs (Gemini) using prompt engineering, without training custom machine learning models.
- The system is implemented as a web-based application and does not support standalone mobile platforms.
- Deployment is limited to a localhost environment for demonstration and evaluation purposes.

Data Constraints:

- AI-assisted functionalities depend on data provided by students and lecturers.

- The quality of AI-generated outputs is influenced by the completeness and clarity of submitted reports and progress updates.
- All system data is used solely for educational and experimental purposes.

Resource Constraints:

- The development team consists of five student members with limited industry experience.
- The system operates independently and is not integrated with the university's official academic management systems.

8.2. Assumptions

- Users are assumed to have stable Internet connectivity during system usage.
- Students are expected to update project progress and submit reports according to the defined workflow.
- Lecturers are expected to review submissions and provide feedback periodically.
- External AI APIs are assumed to be available and operate reliably within their usage limits.

9. Target User / Stakeholder

9.1. Target User

Table 07. *Target User*

Target User	Description	Benefits
Students	Students undertaking graduation thesis or capstone projects, using the system to propose topics, submit reports, and track project progress.	Provides a centralized platform to manage project topics, milestones, deadlines, submissions, and supervisor feedback. The system helps students clearly understand requirements, plan tasks effectively, receive timely notifications, and leverage AI support to detect early risks of delays, thereby improving project quality and on-time completion.
Supervisors (Lecturers)	Lecturers responsible for supervising, monitoring, and evaluating multiple student projects.	Supports centralized management of activities such as topic approval, milestone planning, submission review, and feedback provision. Visual dashboards and AI-assisted features enable supervisors to quickly assess project status, reduce manual workload, and focus more on

		academic guidance.
System Administrators	Personnel responsible for system operation, user account management, and role-based access control.	Enables efficient management of users, roles, and project data in a unified system. Supports system monitoring and ensures data accuracy, security, and stable system operation.

9.2. Stakeholder

Table 08. *Stakeholder*

Stakeholder	Description	Role / Interest
Development Team	A group responsible for analyzing requirements, designing, developing, and deploying the MentorAI Grad system.	Focused on building a system that meets user needs, operates reliably, is scalable, and demonstrates software development capabilities, including the integration of AI-assisted features for graduation project management.

10. Technology Stack

Frontend:

- ReactJS - Used to build web interfaces, dashboards, and progress visualization.
- HTML, CSS, JavaScript - Core technologies for user interface development.

Backend:

- Node.js – Used to implement system logic and develop RESTful APIs.
- JWT (JSON Web Token) – Used for user authentication and authorization.
- bcrypt – Used for secure password hashing.

Database:

- MySQL – Used to store structured data such as user information, project details, milestones, submissions, feedback, and file metadata.
- Uploaded documents (PDF/DOCX) are stored on server or cloud storage, while only file URLs and metadata are stored in the database.

AI & Machine Learning:

- Gemini API – Used to provide AI-assisted summarization of lecturer feedback and student reports through prompt engineering.
- The AI component functions as a supporting tool and does not involve custom model training, predictive analytics, or automated decision-making.

Development Tools:

- GitHub - Used for source code management and team collaboration

- Postman - Used for API testing.
- Figma - Used for UI/UX design.

11. Methodology & Development Plan

11.1. Methodology

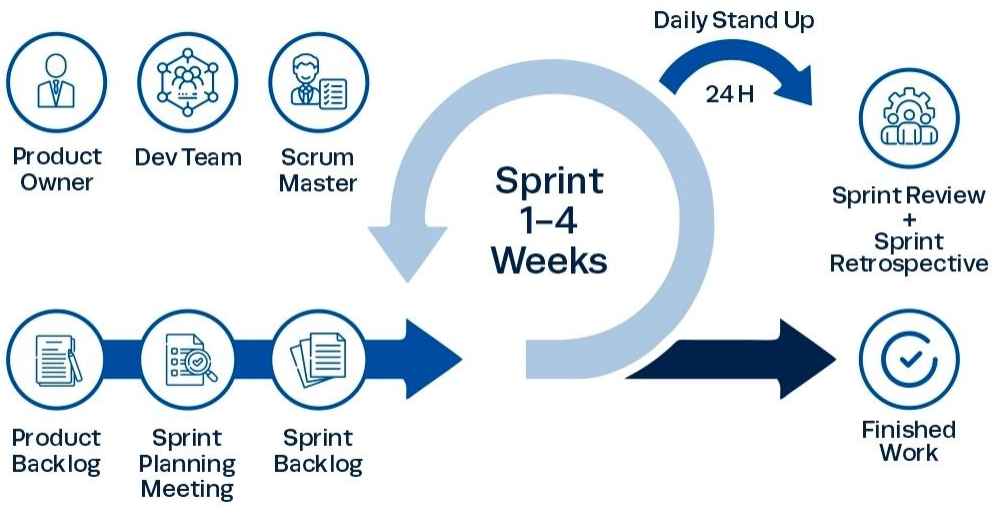

Figure 01. *Scrum Process*

Scrum is an agile project management framework that helps teams structure and manage their work through a set of values, principles, and practices. Much like a rugby team (where it gets its name) training for the big game, scrum encourages teams to learn through experiences, self-organize while working on a problem, and reflect on their wins and losses to continuously improve.

Benefits of the Scrum:

- Adapts Quickly to Change: Scrum allows the team to flexibly adjust plans and goals during short work cycles (Sprints). This helps the project easily react to changes from the client, the market, or technical factors without causing major disruptions.
- Identifies Problems Early: Daily meetings (Daily Scrum) and end-of-Sprint reviews (Sprint Review) provide opportunities for the team to detect and resolve problems early. This helps reduce bug-fixing costs, prevents delays, and improves product quality.
- Focuses on Value: Scrum prioritizes developing the features with the highest value for the client first. The team doesn't waste time on less important requirements, ensuring the final product delivers maximum benefits.

- Better Aligns with Customer Needs: With frequent client involvement and regular Sprint Reviews, the team can receive direct feedback. This helps ensure the product is always adjusted to meet user expectations accurately.
- Increases Work Productivity: Scrum fosters a transparent and collaborative work environment. Team members know their tasks, share information continuously, and therefore work more efficiently and at a faster pace.
- Predictable Scheduling: Fixed-length Sprints help maintain a stable work rhythm. Clients and stakeholders can easily predict progress and plan feature releases, creating consistency and reliability for the entire project.

11.2. Development Plan

11.2.1. Timeline

Table 09. *Timeline*

No.	Task Name	Duration	Start Date	End Date
		100 days	1/2/2026	11/5/2026
1.	Project Initiation	9 days	1/2/2026	10/2/2026
1.1	Requirement Gathering	3 days	1/2/2026	3/2/2026
1.2	Create Proposal Document	6 days	4/2/2026	10/2/2026
2.	Project Setup	18 days	11/2/2026	28/2/2026
2.1	Project Kick-off Meeting	1 days	11/2/2026	11/2/2026
2.2	System Design & Documentation	17 days	12/2/2026	28/2/2026
3.	Development Phase	63 days	1/3/2026	2/5/2026
3.1	Sprint 1 – Core Features Development	21 days	1/3/2026	21/3/2026
3.2	Sprint 2 – Feature Enhancement & Integration	21 days	22/3/2026	11/4/2026
3.3	Sprint 3 – Final Development, Testing & UAT	21 days	12/4/2026	2/5/2026
4.	Project Closure	9 days	3/5/2026	11/5/2026
4.1	Release	4 days	3/5/2026	6/5/2026
4.2	Project Meeting	1 days	7/5/2026	7/5/2026
4.3	Final Submission	4 days	8/5/2026	11/5/2026

11.2.2. Project team

Table 10. *Project team*

Full name	Role	Responsibilities
Thuan Nguyen Trung		Mentor
Tran Dai Nghia	Team member	Frontend Dev, Backend Dev
Tran Van Luc	Team member	Frontend Dev, Backend Dev
Nguyen Thi Nga	Team member	Frontend Dev, Backend Dev
Tu Pham Thi Suong Tuyet	Scrum Master	Frontend Dev, Backend Dev
Tran Anh Thai	Product owner	Frontend Dev, Backend Dev

12. System Architecture Overview

12.1. System Context Diagram

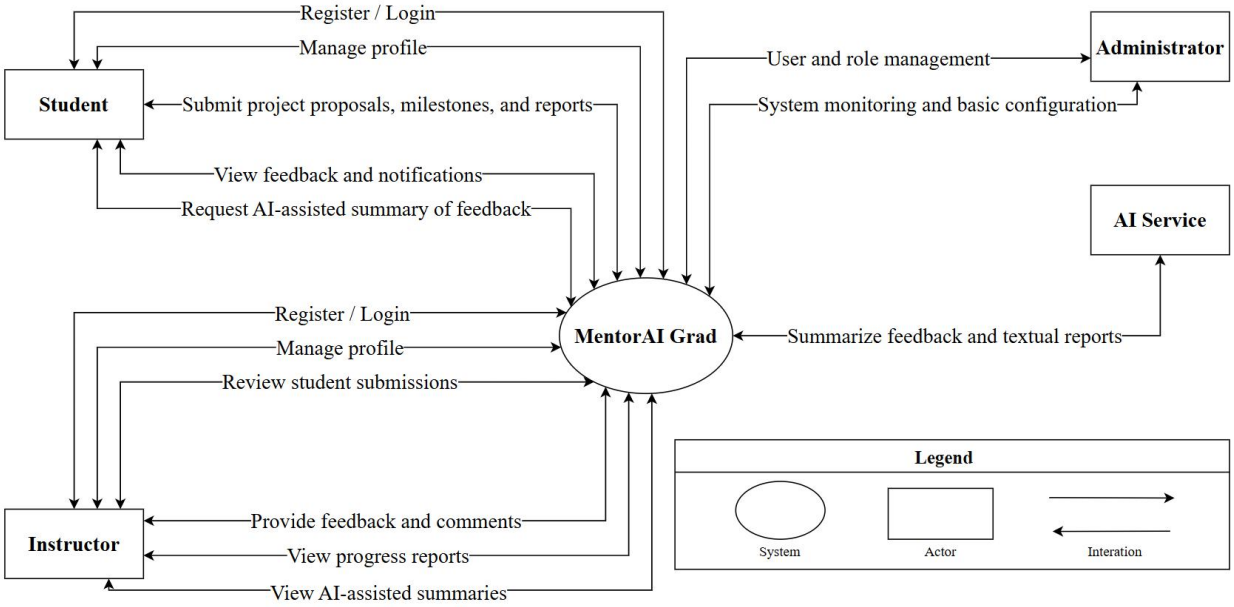

Figure 02. *System Context Diagram*

12.2. System Context Description

Actor: Student

- Registration and Login: Allows students to create accounts and log into the system to access capstone project management functionalities.
- Profile Management: Enables students to update and manage their personal profile information within the system.
- Project Submission: Allows students to submit capstone project proposals, milestone updates, and progress reports according to the defined workflow.
- Feedback and Notification Viewing: Enables students to view feedback, comments from instructors, and system notifications related to their submissions.
- AI-assisted Feedback Summarization: Allows students to request an AI-assisted summary of instructor feedback to improve readability and understanding. This feature is optional and user-initiated.

Actor: Instructor

- Registration and Login: Allows instructors to create accounts and log into the system to manage and supervise assigned capstone projects.
- Profile Management: Enables instructors to manage and update their personal profile information.

- Submission Review and Feedback: Allows instructors to review student submissions, provide feedback, and add comments to support student progress.
- Progress Report Viewing: Enables instructors to view student progress reports and related submissions.
- Instructor – AI-assisted Summary Viewing: Allows instructors to view AI-assisted summaries of student reports and feedback to support supervision activities.

Actor: Administrator

- User and Role Management: Allows administrators to manage user accounts, assign roles (student, instructor, administrator), and control access permissions within the system.
- System Monitoring and Configuration: Enables administrators to monitor basic system operations and perform essential configuration tasks to ensure stable system performance.

Actor: AI Service

- Feedback and Report Summarization: The system integrates with an external AI service (Gemini API) to summarize textual feedback and reports upon request. The AI service functions as a supporting component and does not perform predictive analysis, evaluation, or automated decision-making.

13. Potential Risks and Mitigation Strategies

Table 11. *Potential Risks and Mitigation Strategies*

No.	Risk	Description	Probability (1–5)	Impact (1–5)	Exposure	Mitigation Strategy
1	AI Accuracy Risk	AI may generate inaccurate report summaries or wrongly identify students at risk of delay.	3	4	12	Use AI as a support tool only; supervisors make final decisions. Continuously refine prompts and test AI outputs with sample data.
2	Data Input Risk	Students may not submit reports or	4	4	16	Implement automatic reminders, dashboard

		update milestones on time, leading to incomplete data for tracking and AI analysis.				alerts, and milestone monitoring. Encourage supervisors to enforce submission discipline.
3	System Performance Risk	System may slow down during peak usage or large file uploads.	3	3	9	Optimize database queries, compress uploaded files, limit file sizes, and improve frontend performance.
4	Security Risk	Unauthorized access to student data, reports, or feedback.	2	5	10	Enforce HTTPS, secure authentication, password hashing, and role-based access control. Restrict file access by user roles.
5	External AI Service Risk	AI APIs (OpenAI/Gemini/Claude) may be unavailable or rate-limited.	3	3	9	Ensure core functions work without AI. Provide fallback messages and cache AI results where possible.
6	Schedule Risk	Limited academic timeline may prevent completion of all planned features.	4	3	12	Prioritize MVP features (authentication, milestones, submissions, feedback). Implement AI features after core stability using iterative development.
7	Team	Developmen	3	3	9	Divide tasks

	Experience Risk	t team has limited industry experience.				clearly, use Git for version control, follow documentation, and conduct weekly progress reviews.
8	User Adoption Risk	Students or lecturers may resist switching from manual methods.	2	4	8	Design a simple, intuitive UI; provide basic guidance; focus on solving real issues like deadline tracking and progress visibility.

14. Expected Outcomes / Deliverables

At the end of the project, the following outcomes and deliverables are expected:

14.1. System Deliverables

Fully Functional Web-Based System

- A working capstone project management system accessible through web browsers.
- Role-based access control for students, instructors, and administrators.

Core Functional Modules

- User authentication and profile management
- Project topic and milestone management
- Submission and feedback management
- Basic progress tracking dashboard for monitoring project status

AI-Assisted Features

- AI-assisted feedback summarization to support students in understanding instructor comments and feedback.

Database System

- A centralized database storing user information, project topics, milestones, submissions, feedback, and file references.

14.2. Technical Deliverables

- Source code for frontend (ReactJS) and backend services
- Database schema and design documentation
- API integration documentation for the AI summarization service
- System architecture documentation

14.3. Documentation Deliverables

- Project Proposal and Software Requirements Specification (SRS)
- System Design Document (architecture, database design, diagrams)
- User Manual (for students and instructors)
- Installation and Deployment Guide
- Final Project Report

14.4. Academic & Practical Outcomes

- Demonstration of how AI can support (but not replace) academic supervision through feedback summarization
- Improved transparency in tracking student capstone project progress
- Reduced risk of delayed submissions through structured milestone tracking
- Practical experience for the development team in:
  + Web application development
  + Requirement engineering
  + Integrating AI APIs into real-world systems
  + Team-based software development practices

15. Project Constraint

Table 12. *Project Constraint*

Factor	Condition Description	Condition Description (Project Impact)
Political	Educational policies encourage digital transformation in academic management and supervision activities.	The project aligns with this direction by providing a digital tool for capstone project management. However, the system is implemented as a pilot MVP for Capstone Project 2 and does not aim for institution-wide deployment.
Economic	Financial resources of students and universities are limited, especially for academic projects.	The project should utilize free-tier AI services and local deployment to minimize costs.

	Costs related to AI APIs and server infrastructure must be considered.	The system scope is kept small and suitable for academic conditions.
Social	Students and lecturers are becoming more familiar with online learning, but some still rely on manual management methods (email, spreadsheets). Time pressure and workload make progress tracking difficult.	The system increases transparency and reduces the burden of progress management. A user-friendly interface is essential to encourage adoption and behavior change.
Technological	Rapid development of web technologies, cloud computing, and AI enables the creation of intelligent educational support systems. AI APIs are now easier to integrate into web applications.	This strongly supports the project. However, the system depends on AI API stability and Internet connectivity, so fallback mechanisms are necessary.
Legal	Student data and academic documents are sensitive information. Data privacy and security regulations must be considered.	The system must implement secure authentication, role-based access control, and data protection. AI is used only for support, not for official academic evaluation.
Environmental	Software projects have minimal environmental impact, mainly related to server resource usage and energy consumption.	Using an online system reduces paper usage and printed materials, supporting a more sustainable academic environment.

16. References

Table 13. *References*

No.	References	Document Information
1	Scrum Model	https://en.wikipedia.org/wiki/Scrum_(software_development)
		https://www.scrum.org/resources/scrum-guide
		https://www.atlassian.com/agile/scrum
2	Technical	https://nodejs.org/docs/latest/api/documentation.html
		https://dev.mysql.com/doc/
		https://reactjs.org/docs/getting-started.html
3	Standard	https://www.nws.noaa.gov/oh/hrl/developers_docs/General_Software_Standards.pdf
		https://standards.ieee.org/standard/12208-2017.html